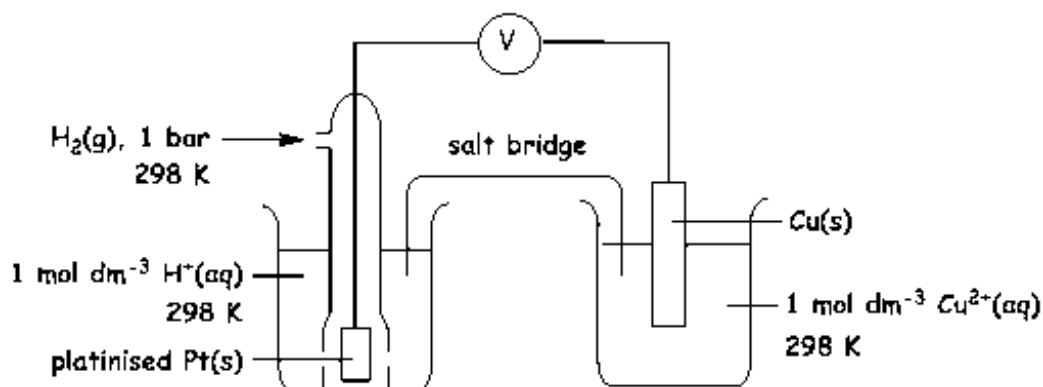


1 (a)



- (b) From Data Booklet:
- | | |
|--|---------|
| $\text{Cu}^{2+} + 2\text{e}^- = \text{Cu}$ | +0.34 V |
| $\text{Ag}^+ + \text{e}^- = \text{Ag}$ | +0.80 V |
| $\text{Zn}^{2+} + 2\text{e}^- = \text{Zn}$ | -0.76 V |

At the anode, Cu dissolves.

Since $E(\text{Zn}^{2+}/\text{Zn})$ is less positive than $E(\text{Cu}^{2+}/\text{Cu})$, zinc impurities dissolve into solution.

Since $E(\text{Ag}^+/\text{Ag})$ is more positive than $E(\text{Cu}^{2+}/\text{Cu})$, the silver impurities will not dissolve into solution and will be collected as anode sludge.

- (c) At the cathode, Cu is deposited.

Since $E(\text{Zn}^{2+}/\text{Zn})$ is less positive than $E(\text{Cu}^{2+}/\text{Cu})$, the zinc impurities will not be oxidised and will thus remain in solution.

- (d) amount of Cu to be coated = $0.50 \div 63.5 = 7.87 \times 10^{-3} \text{ mol}$

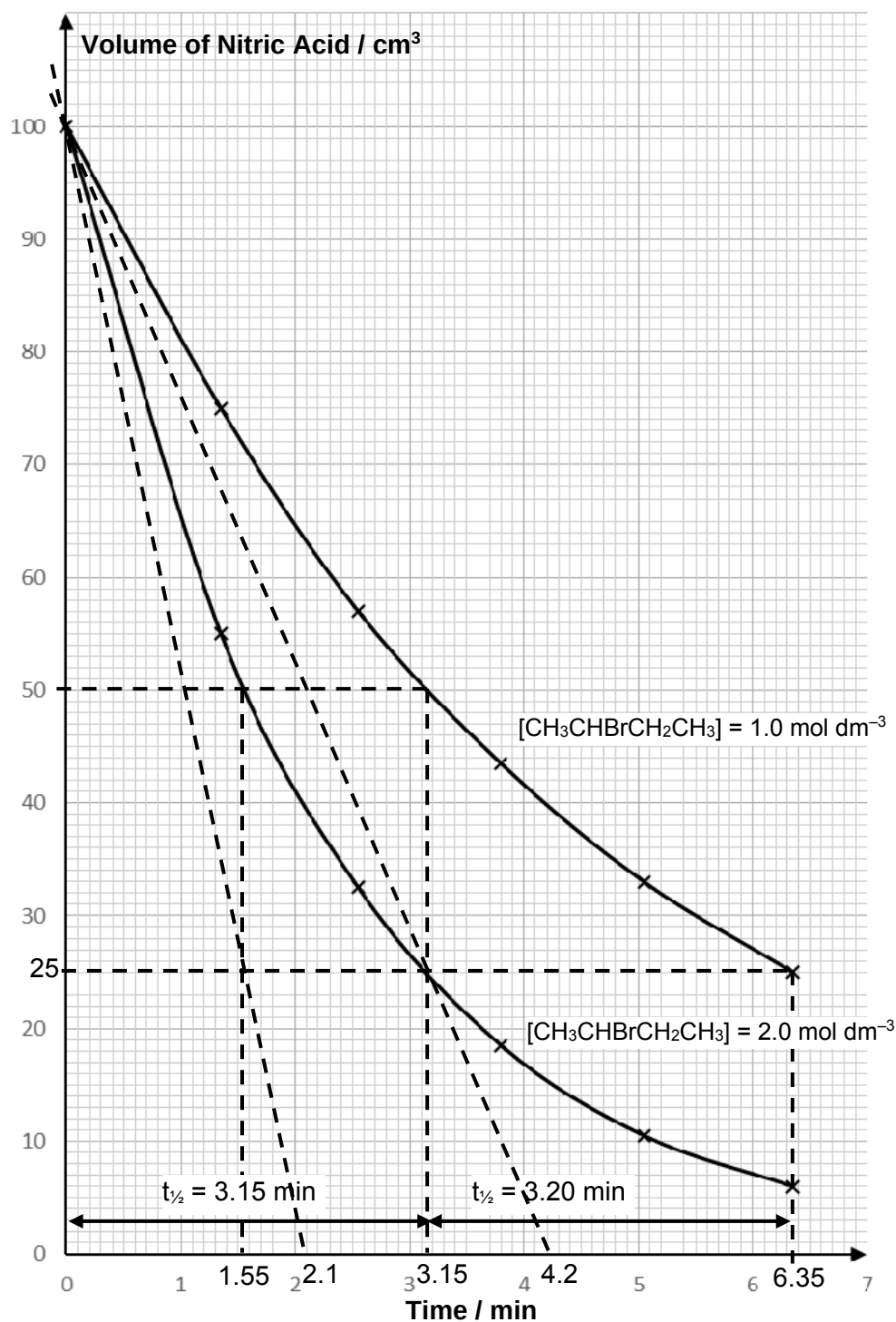
Since $2\text{e}^- \equiv 1\text{Cu}$,

amount of electrons required = $2 \times (7.87 \times 10^{-3}) = 0.0157 \text{ mol}$

Using $Q = It = n_e F$,

$$\text{time taken, } t = \frac{0.0157 \times 96500}{0.20} = 7600 \text{ s}$$

2 (a) (i)



$$\text{average } t_{1/2} = \frac{3.15 + 3.20}{2} = 3.18 \text{ min}$$

Since half-lives are approximately constant, the reaction is first order w.r.t $[\text{NaOH(aq)}]$.

2 (a) (i) When $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3] = 2.0 \text{ mol dm}^{-3}$,

$t_{1/2}$ of NaOH = 1.55 min

When $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3]$ is doubled, half-life of NaOH is (approximately) halved, implying that the rate is (approximately) doubled.

Order of reaction w.r.t $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3]$ is one.

OR

When $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3] = 1.0 \text{ mol dm}^{-3}$,

$$\text{initial rate} = |\text{tangent at } t = 0| = \left| \frac{100 - 0}{0 - 4.2} \right| = 23.8$$

When $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3] = 2.0 \text{ mol dm}^{-3}$,

$$\text{initial rate} = |\text{tangent at } t = 0| = \left| \frac{100 - 0}{0 - 2.1} \right| = 47.6$$

When $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3]$ is doubled, initial rate doubles.

Order of reaction w.r.t $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3]$ is one.

(iii) $\text{rate} = k[\text{CH}_3\text{CHBrCH}_2\text{CH}_3][\text{OH}^-]$

(iv) Since $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3]$ is in large excess as compared to $[\text{NaOH}]$, it can be assumed that $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3]$ remains approximately constant throughout.

$$\therefore \text{rate} = k' [\text{NaOH}] \quad \text{where } k' = k[\text{CH}_3\text{CHBrCH}_2\text{CH}_3]$$

$$t_{1/2} \text{ of NaOH} = \frac{\ln 2}{k'} = \frac{\ln 2}{k[\text{CH}_3\text{CHBrCH}_2\text{CH}_3]}$$

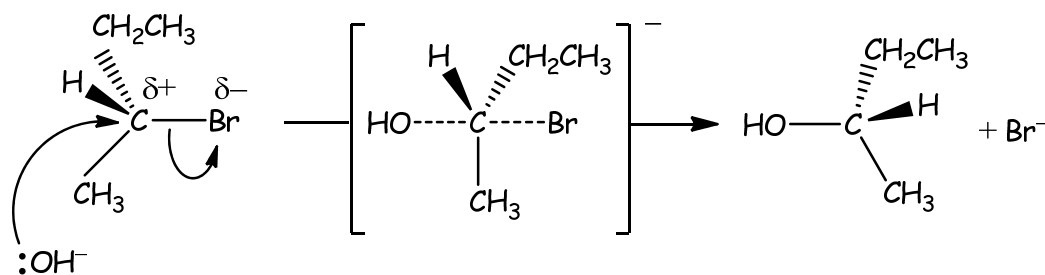
$\therefore t_{1/2}$ of NaOH is independent of $[\text{NaOH}]$ and is 3.18 min.

(v) Using $\frac{\text{final amount}}{\text{initial amount}} = \left(\frac{1}{2}\right)^n$ where n is the no. of half-life passed,

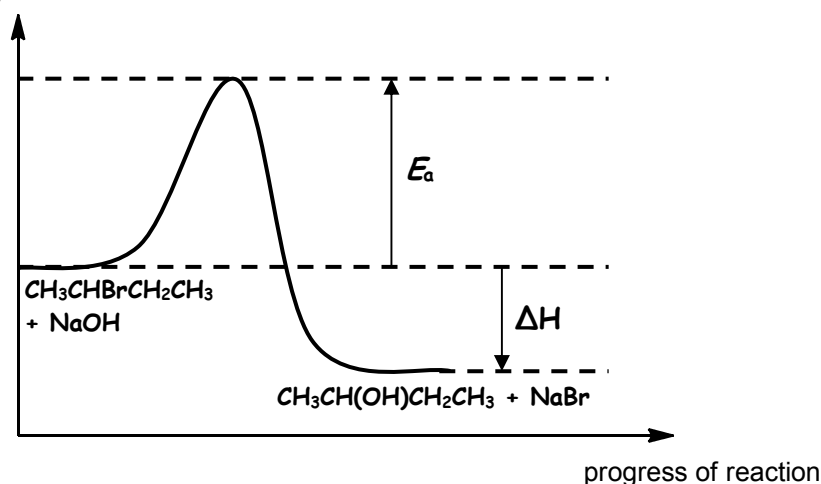
$$\frac{1}{100} = \left(\frac{1}{2}\right)^n \Rightarrow n = 6.64$$

$$\text{time taken} = 6.64 \times 3.18 = 21.1 \text{ min}$$

(vi) Type of mechanism: nucleophilic substitution



2 (b) (ii) Energy / kJ mol^{-1}



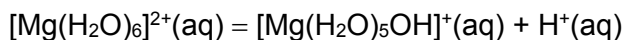
(iii) The rate of reaction will be slower.

Since $E(\text{C}-\text{Cl})$ is larger than $E(\text{C}-\text{Br})$, it implies that $\text{C}-\text{Cl}$ bond is stronger and harder to break compared $\text{C}-\text{Br}$ bond.

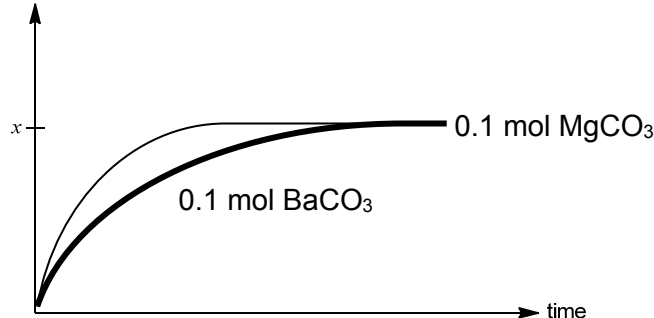
(iv) Yes. Since there is an inversion of stereochemical configuration due to the backside attack of the nucleophile, only one of the enantiomers is produced.

3 (a) A: MgO B: CO₂ C: Mg(OH)₂ D: MgSO₄ E: CaCO₃

(b) MgSO₄ dissolves in water to give $[\text{Mg}(\text{H}_2\text{O})_6]^{2+}$, which hydrolyses slightly due to slightly higher charge density of Mg^{2+} to give an acidic solution.



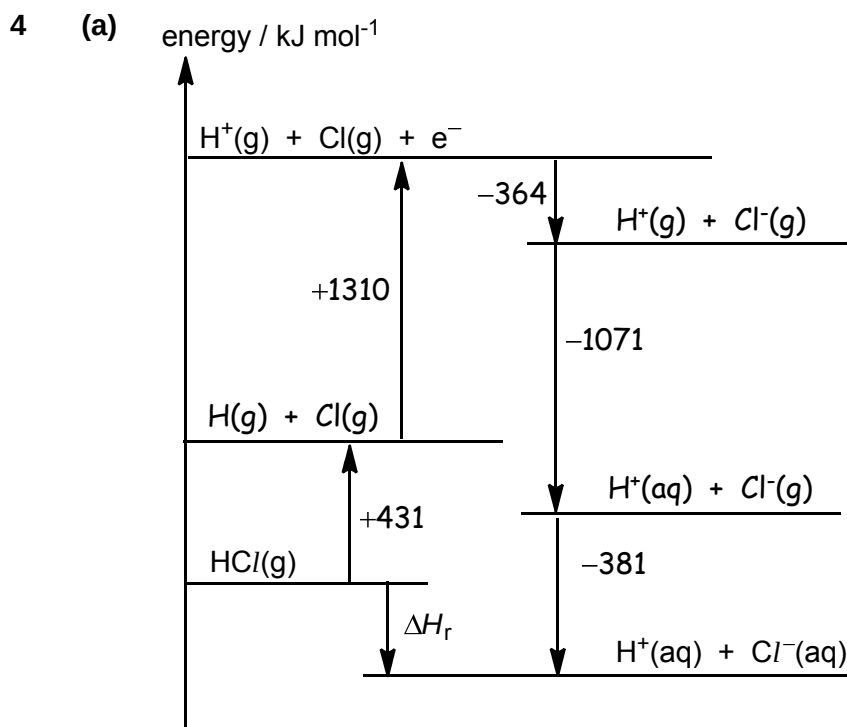
(c) (i) Volume of gas B



(ii) Since $1 \text{ CO}_2 \equiv 1 \text{ MCO}_3$ and amount of BaCO₃ used is the same, the total volume of gas B formed is the same.

Compared to Mg^{2+} , Ba^{2+} has a larger radius and smaller charge density.

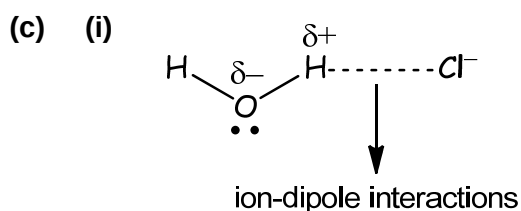
Hence, Ba^{2+} is less able to polarise large CO_3^{2-} anion and BaCO₃ is more thermally stable and decomposes less readily, resulting in a longer heating time.



By Hess's Law,

$$\Delta H_r = (+431) + (+1310) + (-364) + (-1071) + (-381) = -75.0 \text{ kJ mol}^{-1}$$

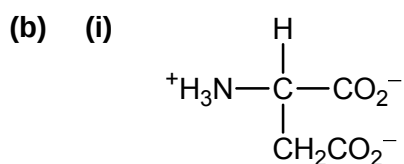
(b) $\Delta G_{\text{soln}} = -75 - (-18) = -57.0 \text{ kJ mol}^{-1}$



(ii) Compared to Cl^- , Br^- has larger radius and lower charge density, resulting in a weaker ion-dipole interactions formed between Br^- and H_2O molecule.
 $\therefore \text{Br}^-$ has a smaller magnitude of ΔH_{hyd} .

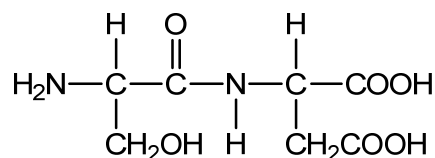
(e) Test : Add $\text{Cl}_2(\text{aq})$ to both samples separately.
 Observation : For $\text{HBr}(\text{aq})$, colourless solution turns orange, due to $\text{Br}_2(\text{aq})$.
 For $\text{HCl}(\text{aq})$, colourless solution remains.

5 (a) $\text{NaOH}(\text{aq})$ (or $\text{KOH}(\text{aq})$ or $\text{H}_2\text{SO}_4(\text{aq})$ or $\text{HCl}(\text{aq})$), heat for prolonged period.



(ii) anode

5 (c) (i)



(ii) ser-asp-tyr-val-gly-ser

reagent & condition	Aqueous Br ₂	Cold aqueous NaOH	CH ₃ COOH
product	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_3\text{Br}_2\text{OH} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COO}^-\text{Na}^+ \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_5\text{O}^-\text{Na}^+ \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_5\text{OH} \\ \text{CH}_3\text{COO}^- \end{array}$

6 (a) (i) amount of HCl used = $0.10 \times \frac{30.0}{1000} = 3.00 \times 10^{-3} \text{ mol}$

Since 1 HCl $\equiv$ 1 NaOH,

amount of NaOH left unreacted = $3.00 \times 10^{-3} \text{ mol}$

total amount of NaOH added initially = $0.10 \times \frac{50.0}{1000} = 5.00 \times 10^{-3} \text{ mol}$

amount of NaOH reacted with Br₂ = $(5.00 \times 10^{-3}) - (3.00 \times 10^{-3})$
= $2.00 \times 10^{-3} \text{ mol}$

amount of Br₂ used = $0.10 \times \frac{10.0}{1000} = 1.00 \times 10^{-3} \text{ mol}$

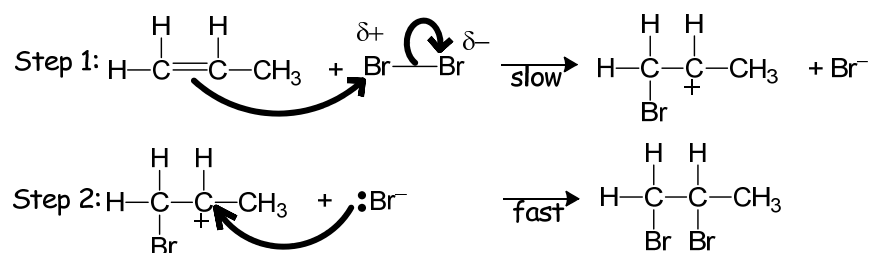
$\therefore$ Mole ratio of Br₂ : NaOH = $1.00 \times 10^{-3} : 2.00 \times 10^{-3} = 1 : 2$

(ii) $\text{Br}_2 + 2\text{OH}^- \rightarrow \text{Br}^- + \text{BrO}^- + \text{H}_2\text{O}$
value of x: 1

(b) (i) CH₂BrCHBrCH₃

(ii) ⁷⁹Br and ⁸¹Br in the ratio of 1 : 1

(iii) type of mechanism: electrophilic addition



- 6 (c) (i) Since the O atom of H_2O also possesses a lone pair of electrons, the O atom of H_2O can act as a nucleophile to attack the electron-deficient C^+ of carbocation, forming the halohydrin.

Since water is present in large excess compared to Br_2 , there will be relatively much more H_2O molecule to act as the nucleophile than the Br^- present in the mixture.

